# MT-104
# Linear Algebra

FALL 2024

**Muhammad Ali**

*m.ali@nu.edu.pk*

National University of Computer and Emerging Sciences

October 6, 2024

# Section 1.8

## Introduction to Linear Transformations

# Motivation

Let $A$ be an $m \times n$ matrix. For the matrix equation $Ax = b$ we have learned to describe

- the solution set: all $x$ in $\mathbf{R}^n$ making the equation true.
- the column span: the set of all $b$ in $\mathbf{R}^m$ making the equation consistent.

It turns out these two sets are very closely related to each other.

In order to understand this relationship, it helps to think of the matrix $A$ as a *transformation* from $\mathbf{R}^n$ to $\mathbf{R}^m$.

It's a special kind of transformation called a *linear transformation*.

This is also a way to understand the *geometry* of *matrices*.

# Transformations

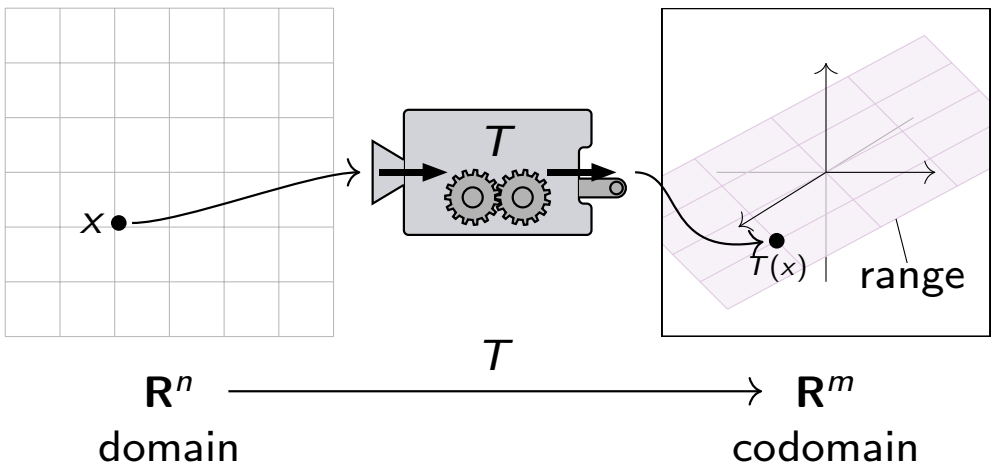
## Definition

A **transformation** (or **function** or **map**) from $\mathbf{R}^n$ to $\mathbf{R}^m$ is a rule $T$ that assigns to each vector $x$ in $\mathbf{R}^n$ a vector $T(x)$ in $\mathbf{R}^m$.

- $\mathbf{R}^n$ is called the **domain** of $T$ (the inputs).
- $\mathbf{R}^m$ is called the **codomain** of $T$ (the outputs).
- For $x$ in $\mathbf{R}^n$, the vector $T(x)$ in $\mathbf{R}^m$ is the **image** of $x$ under $T$.
  Notation: $x \mapsto T(x)$.
- The set of all images $\{T(x) \mid x \text{ in } \mathbf{R}^n\}$ is the **range** of $T$.

Notation:

$T\colon \mathbf{R}^n \longrightarrow \mathbf{R}^m$ means $T$ is a transformation from $\mathbf{R}^n$ to $\mathbf{R}^m$.




It may help to think of $T$ as a "machine" that takes $x$ as an input, and gives you $T(x)$ as the output.

# Functions from Calculus

Many of the functions you know and love have domain and codomain $\mathbf{R}$.

$$\sin : \mathbf{R} \longrightarrow \mathbf{R} \qquad \sin(x) = \begin{pmatrix} \text{the length of the opposite edge over the} \\ \text{hypotenuse of a right triangle with angle} \\ x \text{ in radians} \end{pmatrix}$$

Note how I've written down the *rule* that defines the function sin.

$$f : \mathbf{R} \longrightarrow \mathbf{R} \qquad f(x) = x^2$$

Note that "$x^2$" is sloppy (but common) notation for a function: it doesn't have a name!

You may be used to thinking of a function in terms of its graph.

$(x, \sin x)$

•

•
$x$

The horizontal axis is the domain, and the vertical axis is the codomain.

This is fine when the domain and codomain are $\mathbf{R}$, but it's hard to do when they're $\mathbf{R}^2$ and $\mathbf{R}^3$! You need five dimensions to draw that graph.

# Matrix Transformations

Most of the transformations we encounter in this class will come from (surprise) matrices!

## Definition

Let $A$ be an $m \times n$ matrix. The **matrix transformation** associated to $A$ is the transformation

$$T : \mathbf{R}^n \longrightarrow \mathbf{R}^m \quad \text{defined by} \quad T(x) = Ax.$$

$$A_{m\times n} X_{n\times 1}$$

In other words, $T$ takes the vector $x$ in $\mathbf{R}^n$ to the vector $Ax$ in $\mathbf{R}^m$.

Your life will be much easier if you just remember these.

- The *domain* of $T$ is $\mathbf{R}^n$, which is the number of *columns* of $A$.
- The *codomain* of $T$ is $\mathbf{R}^m$, which is the number of *rows* of $A$.
- The *range* of $T$ is the set of all images of $T$:

$$T(x) = Ax = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 v_1 + x_2 v_2 + \cdots + x_n v_n.$$

This is the *column span* of $A$. It is a span of vectors in the codomain.

# Matrix Transformations

## Example

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^3$.

- If $u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ then $T(u) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \\ 7 \end{pmatrix}$.
- Let $b = \begin{pmatrix} 7 \\ 5 \\ 7 \end{pmatrix}$. Find $v$ in $\mathbf{R}^2$ such that $T(v) = b$. Is there more than one?

  We want to find $v$ such that $T(v) = Av = b$. We know how to do that:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} v = \begin{pmatrix} 7 \\ 5 \\ 7 \end{pmatrix} \xrightarrow{\text{augmented matrix}} \left(\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & 1 & 5 \\ 1 & 1 & 7 \end{array}\right) \xrightarrow{\text{row reduce}} \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{array}\right).$$

  This gives $x = 2$ and $y = 5$, or $v = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ (unique). In other words,

$$T(v) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 7 \\ 5 \\ 7 \end{pmatrix}.$$

# Matrix Transformations

## Example, continued

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^3$.

- Is there any $c$ in $\mathbf{R}^3$ such that there is more than one $v$ in $\mathbf{R}^2$ with $T(v) = c$?

  Translation: is there any $c$ in $\mathbf{R}^3$ such that the solution set of $Ax = c$ has more than one vector $v$ in it?

  The solution set of $Ax = c$ is a translate of the solution set of $Ax = b$ (from before), which has one vector in it. So the solution set to $Ax = c$ has only one vector. So no!

- Find $c$ such that there is *no* $v$ with $T(v) = c$.

  Translation: Find $c$ such that $Ax = c$ is inconsistent.

  Translation: Find $c$ not in the column span of $A$ (i.e., the range of $T$).

  We could draw a picture, or notice: $a\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a+b \\ b \\ a+b \end{pmatrix}$. So anything in the column span has the same first and last coordinate. So $c = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ is not in the column span (for example).

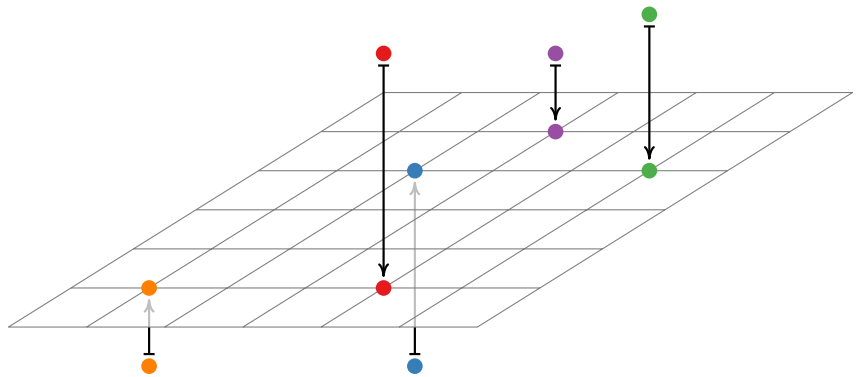
# Matrix Transformations

## Geometric example

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^3 \to \mathbf{R}^3$. Then

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}.$$

This is *projection onto the xy-axis*. Picture:



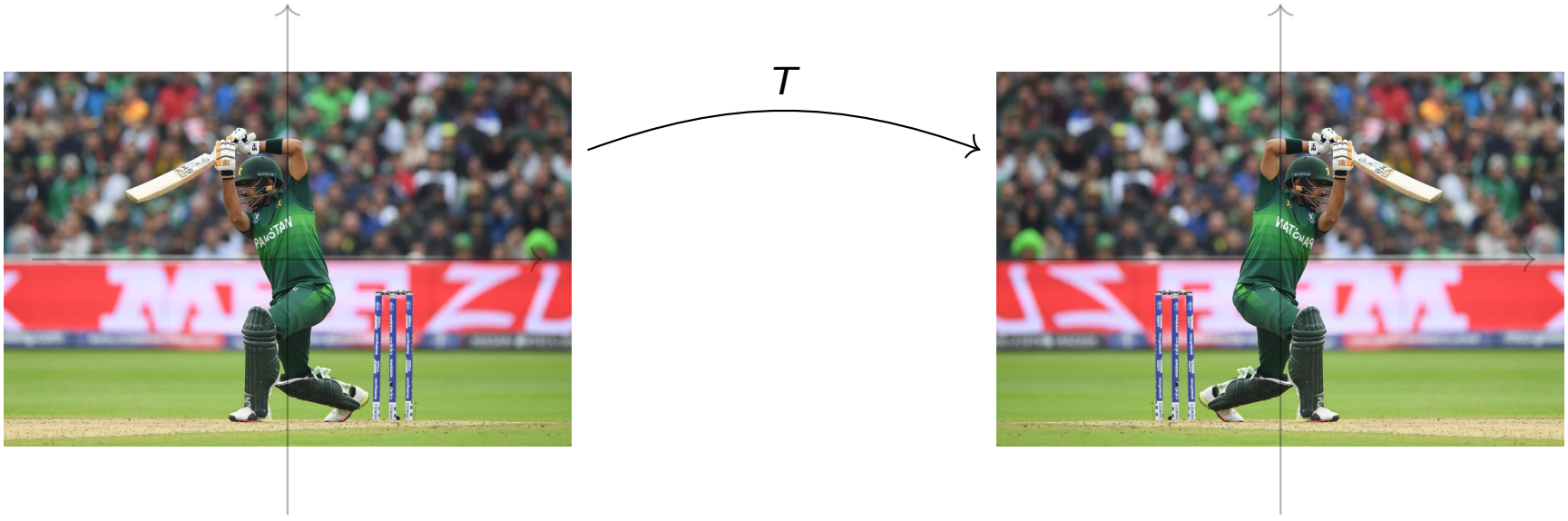
# Matrix Transformations

## Geometric example

Let $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^2$. Then

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}.$$

This is *reflection over the y-axis*. Picture:



# Matrix Transformations

## Geometric example

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^2$. Then

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}.$$

This is *reflection over the x-axis*.

Let $A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^2$. Then

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}.$$

This is *reflection about origin*.

Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^2$. Then

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}.$$

This is *reflection about the line* $y = x$.

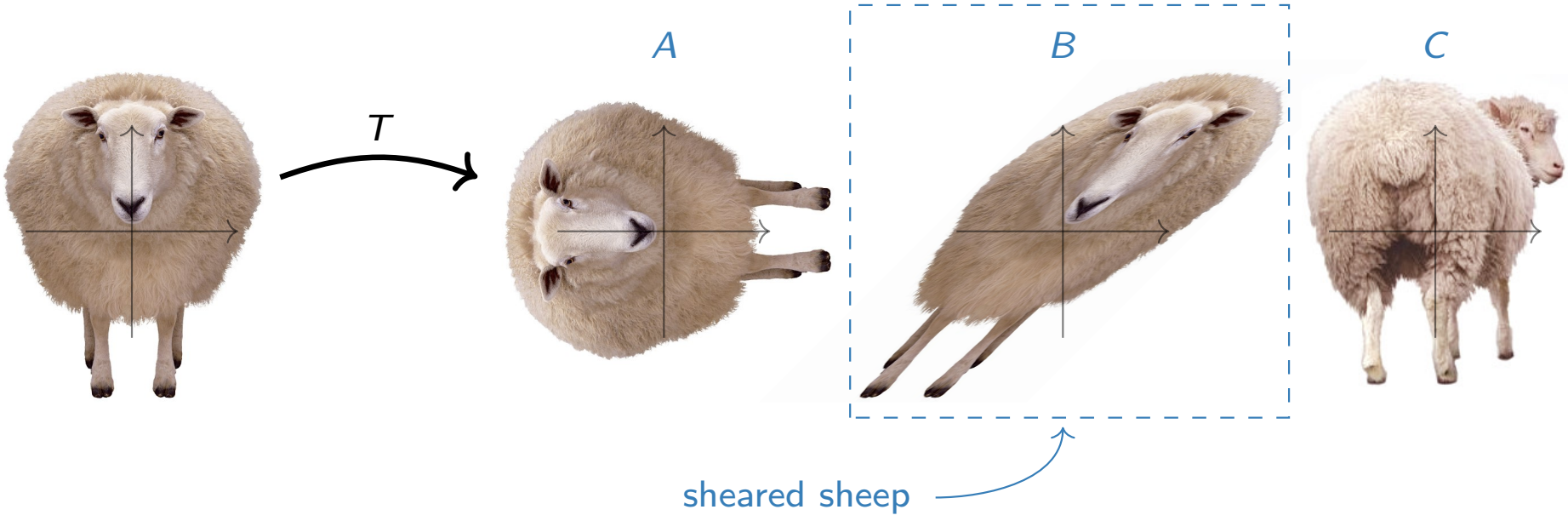
# Poll

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and let $T(x) = Ax$, so $T\colon \mathbf{R}^2 \to \mathbf{R}^2$. ($T$ is called a **shear**.)

**Poll**

What does $T$ do to this sheep?

Hint: first draw a picture what it does to the box *around* the sheep.



# Linear Transformations

Recall: If $A$ is a matrix, $u, v$ are vectors, and $c$ is a scalar, then

$$A(u+v) = Au + Av \qquad A(cv) = cAv.$$

So if $T(x) = Ax$ is a matrix transformation then,

$$T(u+v) = T(u) + T(v) \qquad T(cv) = cT(v).$$

This property is so special that it has its own name.

## Definition

A transformation $T\colon \mathbf{R}^n \to \mathbf{R}^m$ is **linear** if it satisfies the above equations for all vectors $u, v$ in $\mathbf{R}^n$ and all scalars $c$.

In other words, $T$ "respects" addition and scalar multiplication.

Check: if $T$ is linear, then

$$T(0) = 0 \qquad T(cu + dv) = cT(u) + dT(v)$$

for all vectors $u, v$ and scalars $c, d$. More generally,

$$T\left(c_1 v_1 + c_2 v_2 + \cdots + c_n v_n\right) = c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n).$$

In engineering this is called **superposition**.

# Linear Transformations

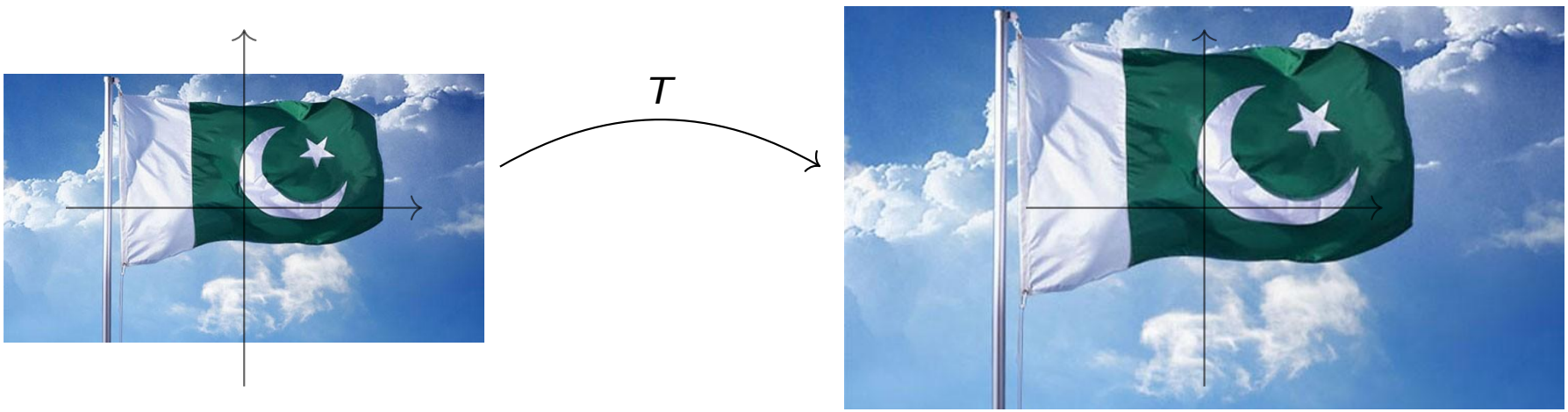
## Dilation

Define $T\colon \mathbf{R}^2 \to \mathbf{R}^2$ by $T(x) = 1.5x$. Is $T$ linear? Check:

$$T(u+v) = 1.5(u+v) = 1.5u + 1.5v = T(u) + T(v)$$
$$T(cv) = 1.5(cv) = c(1.5v) = c(Tv).$$

So $T$ satisfies the two equations, hence $T$ is linear.

This is called **dilation** or **scaling** (by a factor of 1.5). Picture:



# Linear Transformations

## Rotation

Define $T\colon \mathbf{R}^2 \to \mathbf{R}^2$ by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

Is $T$ linear? Check:

$$T\left(\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right) = \begin{pmatrix} -u_2 \\ u_1 \end{pmatrix} + \begin{pmatrix} -v_2 \\ v_1 \end{pmatrix} = \begin{pmatrix} -(u_2+v_2) \\ (u_1+v_1) \end{pmatrix} = T\begin{pmatrix} u_1+u_2 \\ v_1+v_2 \end{pmatrix}$$

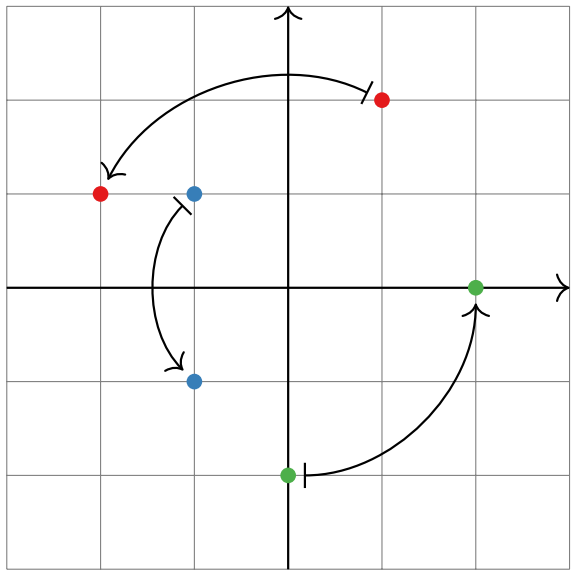
$$T\left(c\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right) = T\begin{pmatrix} cv_1 \\ cv_2 \end{pmatrix} = \begin{pmatrix} -cv_2 \\ cv_1 \end{pmatrix} = c\begin{pmatrix} -v_2 \\ v_1 \end{pmatrix} = cT\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

So $T$ satisfies the two equations, hence $T$ is linear. This is called **rotation** (by 90°). Picture:

$$T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$



# Section 1.9

## The Matrix of a Linear Transformation

# Unit Coordinate Vectors

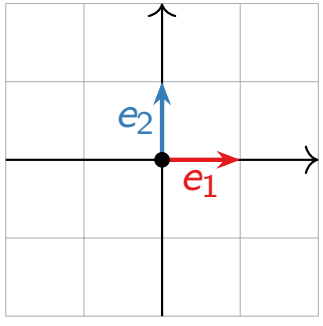
## Definition

The **unit coordinate vectors** in $\mathbf{R}^n$ are

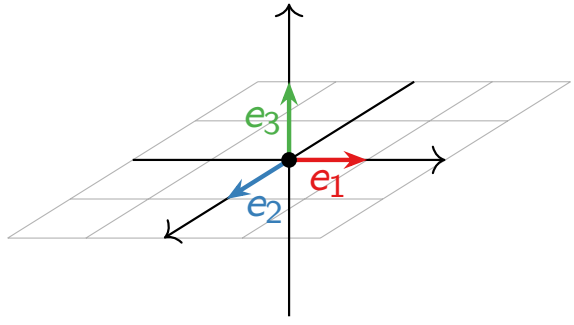
This is what $e_1, e_2, \ldots$ mean, *for the rest of the class.*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \ldots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{pmatrix}, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

in $\mathbf{R}^2$




in $\mathbf{R}^3$




Note: if $A$ is an $m \times n$ matrix with columns $v_1, v_2, \ldots, v_n$, then $Ae_i = v_i$ for $i = 1, 2, \ldots, n$: multiplying a matrix by $e_i$ gives you the $i$th column.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

# Linear Transformations are Matrix Transformations

Recall: A matrix $A$ defines a linear transformation $T$ by $T(x) = Ax$.

**Theorem**

Let $T\colon \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation. Let

$$A = \begin{pmatrix} | & | & & | \\ T(e_1) & T(e_2) & \cdots & T(e_n) \\ | & | & & | \end{pmatrix}.$$

This is an $m \times n$ matrix, and $T$ is the matrix transformation for $A$: $T(x) = Ax$.

The matrix $A$ is called the **standard matrix** for $T$.

**Take-Away**

Linear transformations are the same as matrix transformations.

**Dictionary**

Linear transformation $T\colon \mathbf{R}^n \to \mathbf{R}^m$ $\rightsquigarrow$ $m \times n$ matrix $A = \begin{pmatrix} | & | & & | \\ T(e_1) & T(e_2) & \cdots & T(e_n) \\ | & | & & | \end{pmatrix}$

$T(x) = Ax$, $T\colon \mathbf{R}^n \to \mathbf{R}^m$ $\leftsquigarrow$ $m \times n$ matrix $A$

# Linear Transformations are Matrix Transformations

## Continued

Why is a linear transformation a matrix transformation?

Suppose for simplicity that $T\colon \mathbf{R}^3 \to \mathbf{R}^2$.

$$
\begin{aligned}
T\begin{pmatrix} x \\ y \\ z \end{pmatrix} &= T\left( x\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + z\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) \\
&= T\left(xe_1 + ye_2 + ze_3\right) \\
&= xT(e_1) + yT(e_2) + zT(e_3) \\
&= \begin{pmatrix} | & | & | \\ T(e_1) & T(e_2) & T(e_3) \\ | & | & | \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
&= A\begin{pmatrix} x \\ y \\ z \end{pmatrix}.
\end{aligned}
$$

# Linear Transformations are Matrix Transformations

## Example

Before, we defined a **dilation** transformation $T\colon \mathbf{R}^2 \to \mathbf{R}^2$ by $T(x) = 1.5x$. What is its standard matrix?

$$\left.\begin{aligned} T(e_1) = 1.5e_1 = \begin{pmatrix} 1.5 \\ 0 \end{pmatrix} \\ T(e_2) = 1.5e_2 = \begin{pmatrix} 0 \\ 1.5 \end{pmatrix} \end{aligned}\right\} \Longrightarrow A = \begin{pmatrix} 1.5 & 0 \\ 0 & 1.5 \end{pmatrix}.$$

Check:

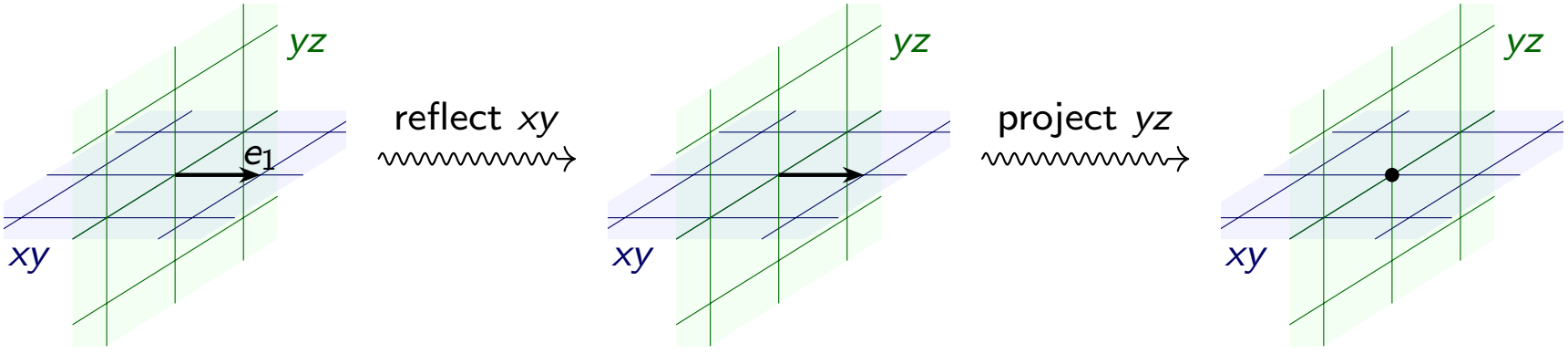
$$\begin{pmatrix} 1.5 & 0 \\ 0 & 1.5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1.5x \\ 1.5y \end{pmatrix} = 1.5 \begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} x \\ y \end{pmatrix}.$$

# Linear Transformations are Matrix Transformations

## Example

### Question

What is the matrix for the linear transformation $T\colon \mathbf{R}^3 \to \mathbf{R}^3$ that reflects through the $xy$-plane and then projects onto the $yz$-plane?



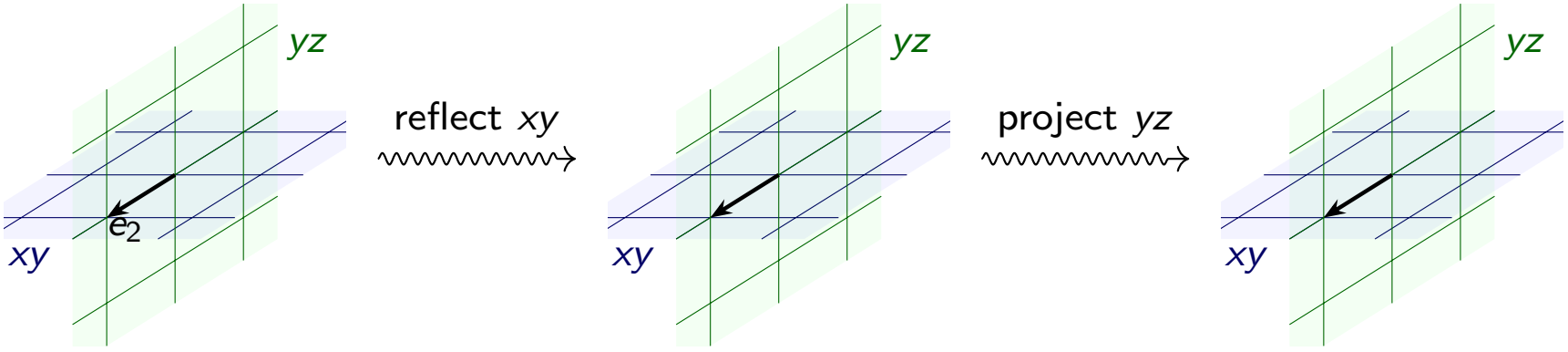

$$T(e_1) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

# Linear Transformations are Matrix Transformations

## Example, continued

### Question

What is the matrix for the linear transformation $T\colon \mathbf{R}^3 \to \mathbf{R}^3$ that reflects through the $xy$-plane and then projects onto the $yz$-plane?



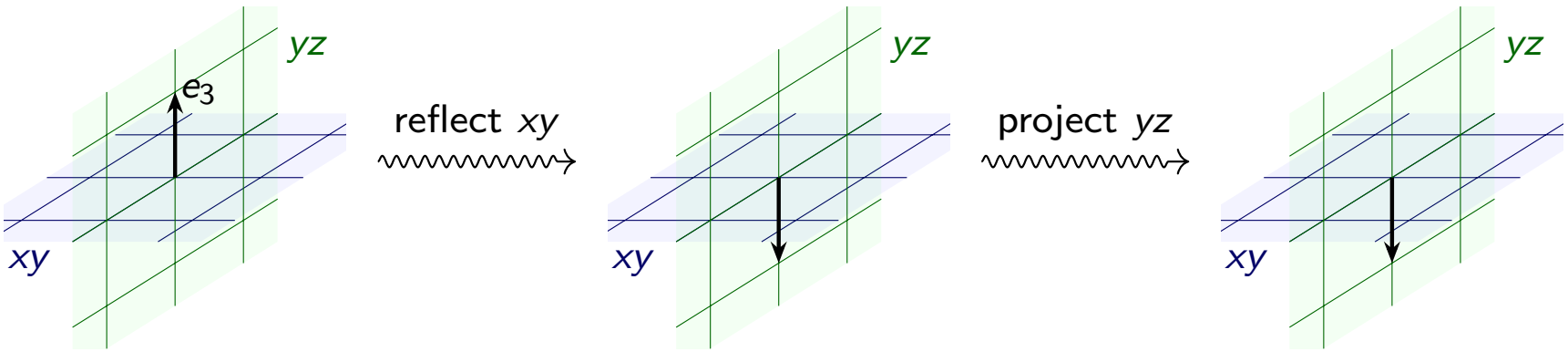

$$T(e_2) = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

# Linear Transformations are Matrix Transformations

## Example, continued

### Question

What is the matrix for the linear transformation $T\colon \mathbf{R}^3 \to \mathbf{R}^3$ that reflects through the $xy$-plane and then projects onto the $yz$-plane?




$$T(e_3) = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}.$$

# Linear Transformations are Matrix Transformations

## Example, continued

### Question

What is the matrix for the linear transformation $T\colon \mathbf{R}^3 \to \mathbf{R}^3$ that reflects through the $xy$-plane and then projects onto the $yz$-plane?

$$\left.\begin{aligned} T(e_1) &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ T(e_2) &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ T(e_1) &= \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \end{aligned}\right\} \implies A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

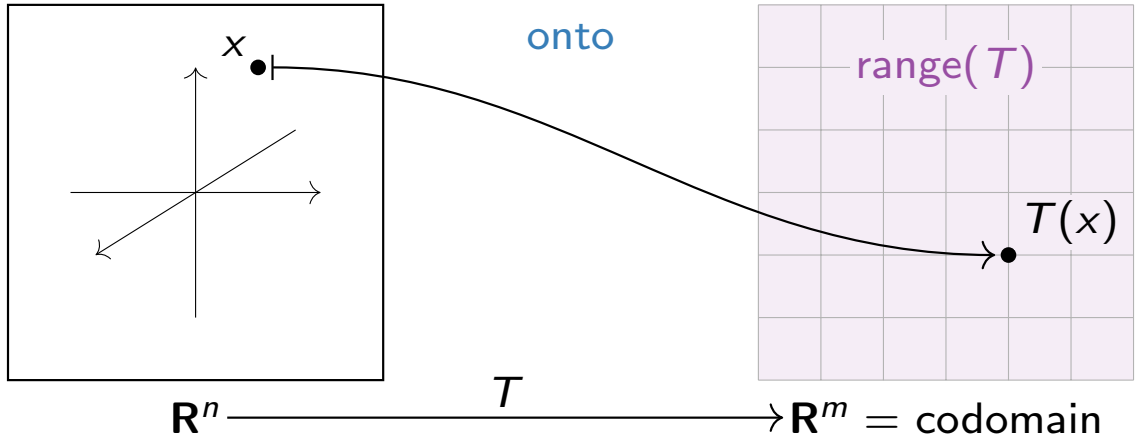
# Other Geometric Transformations

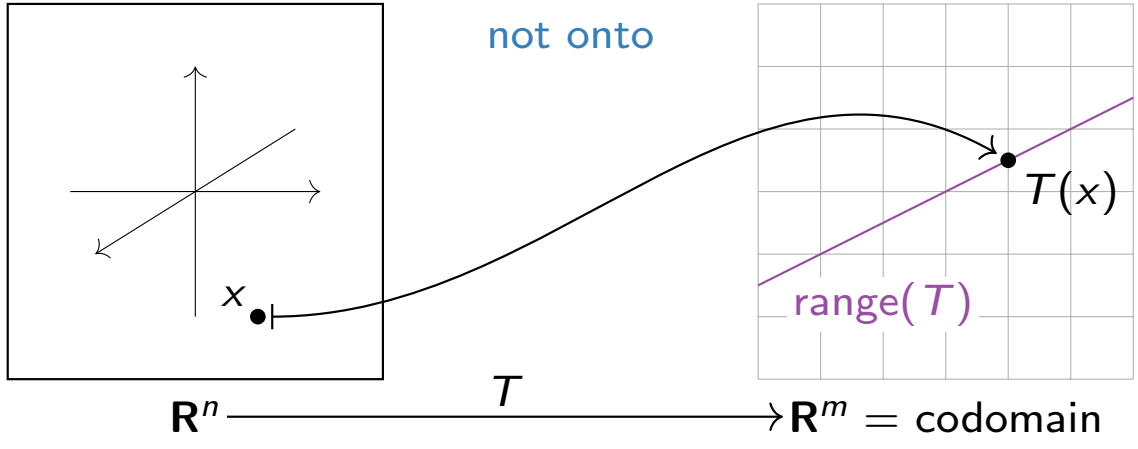
There is a long list of geometric transformations of $\mathbf{R}^2$ in §1.9 of Lay. (Reflections over the diagonal, contractions and expansions along different axes, shears, projections, . . . ) Please look them over.

# Onto Transformations

## Definition

A transformation $T: \mathbf{R}^n \to \mathbf{R}^m$ is **onto** (or **surjective**) if the range of $T$ is equal to $\mathbf{R}^m$ (its codomain). In other words, each $b$ in $\mathbf{R}^m$ is the image of *at least one* $x$ in $\mathbf{R}^n$: every possible output has an input. Note that *not* onto means there is some $b$ in $\mathbf{R}^m$ which is not the image of any $x$ in $\mathbf{R}^n$.



# Characterization of Onto Transformations

## Theorem

Let $T: \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation with matrix $A$. Then the following are equivalent:

- $T$ is onto
- $T(x) = b$ has a solution for every $b$ in $\mathbf{R}^m$
- $Ax = b$ is consistent for every $b$ in $\mathbf{R}^m$
- The columns of $A$ span $\mathbf{R}^m$
- $A$ has a pivot in every row

## Question

If $T: \mathbf{R}^n \to \mathbf{R}^m$ is onto, what can we say about the relative sizes of $n$ and $m$?

Answer: $T$ corresponds to an $m \times n$ matrix $A$. In order for $A$ to have a pivot in every row, it must have *at least as many* columns as rows: $m \leq n$.

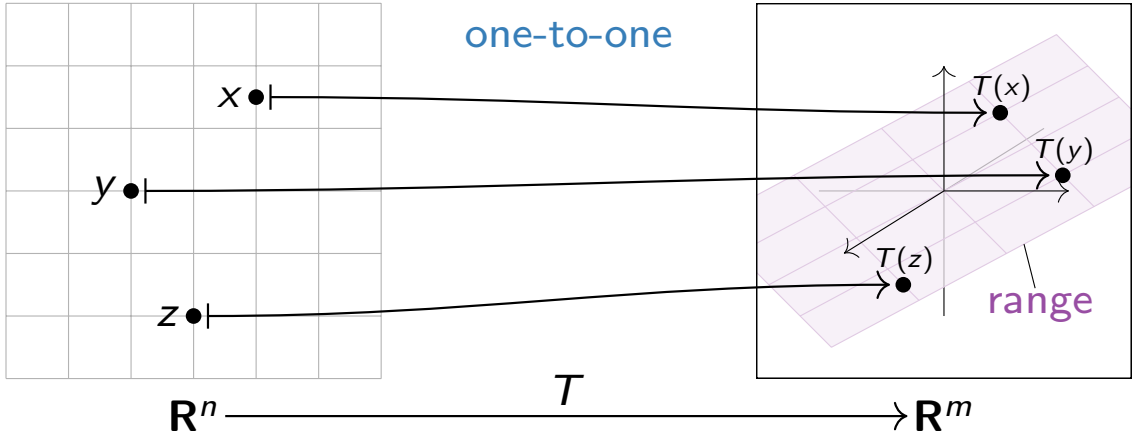
$$\begin{pmatrix} 1 & 0 & \star & 0 & \star \\ 0 & 1 & \star & 0 & \star \\ 0 & 0 & 0 & 1 & \star \end{pmatrix}$$

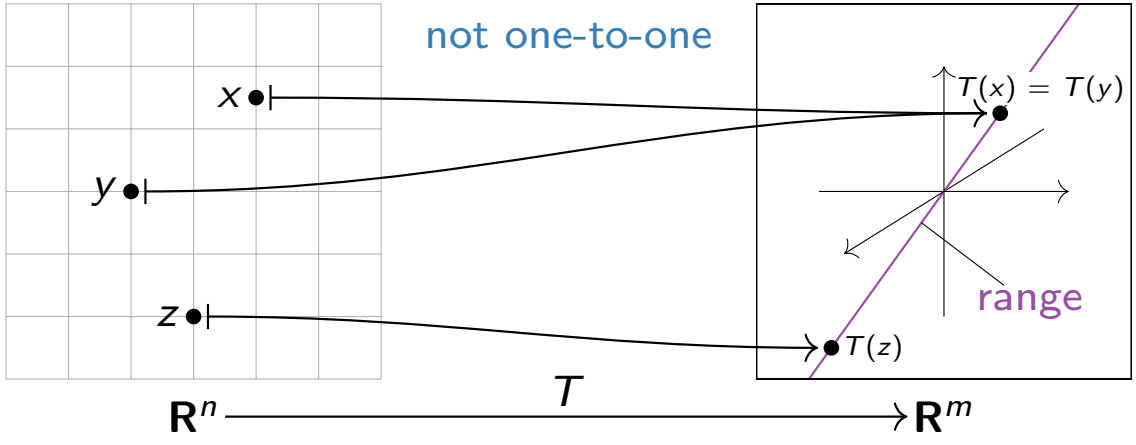
For instance, $\mathbf{R}^2$ is "too small" to map *onto* $\mathbf{R}^3$.

# One-to-one Transformations

## Definition

A transformation $T: \mathbf{R}^n \to \mathbf{R}^m$ is **one-to-one** (or **into**, or **injective**) if different vectors in $\mathbf{R}^n$ map to different vectors in $\mathbf{R}^m$. In other words, each $b$ in $\mathbf{R}^m$ is the image of *at most one* $x$ in $\mathbf{R}^n$: different inputs have different outputs. Note that *not* one-to-one means different vectors in $\mathbf{R}^n$ have the same image.



# Characterization of One-to-One Transformations

## Theorem

Let $T: \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation with matrix $A$. Then the following are equivalent:

- $T$ is one-to-one
- $T(x) = b$ has one or zero solutions for every $b$ in $\mathbf{R}^m$
- $Ax = b$ has a unique solution or is inconsistent for every $b$ in $\mathbf{R}^m$
- $Ax = 0$ has a unique solution
- The columns of $A$ are linearly independent
- $A$ has a pivot in every column.

## Question

If $T: \mathbf{R}^n \to \mathbf{R}^m$ is one-to-one, what can we say about the relative sizes of $n$ and $m$?

Answer: $T$ corresponds to an $m \times n$ matrix $A$. In order for $A$ to have a pivot in every column, it must have *at least as many* rows as columns: $n \leq m$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

For instance, $\mathbf{R}^3$ is "too big" to map *into* $\mathbf{R}^2$.